I2 — Inorganic Chemistry II

1. Properties of Period 3 Oxides

1.1 Types of Bonding in Period 3 Oxides

As you cross Period 3 from left to right (Na to Cl), the bonding in oxides changes from ionic to covalent. This is because electronegativity increases across the period, reducing the electronegativity difference between the element and oxygen. The smaller the difference, the more covalent the bonding character.

Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3	Cl_2O_7
Bonding	Ionic	Ionic	Ionic (some covalent character)	Giant covalent	Simple covalent	Simple covalent	Simple covalent
Structure	Giant ionic	Giant ionic	Giant ionic	Giant covalent (macromolecular)	Simple molecular	Simple molecular	Simple molecular
Melting point	High (1132°C)	Very high (2852°C)	Very high (2072°C)	Very high (1713°C)	Low (340°C)	Low (17°C)	Low (-91°C)
Electrical conductivity	Conducts when molten/dissolved	Conducts when molten	Insulator (too high lattice energy to melt easily)	Insulator	Non-conductor	Non-conductor	Non-conductor

Explaining the trend: Na_2O and MgO are giant ionic lattices held by strong electrostatic forces between ions, giving high melting points. MgO has a higher melting point than Na_2O because Mg^{2+} and O^{2-} have higher charges and smaller ionic radii, producing stronger ionic bonds. Al_2O_3 has high lattice energy due to the 3+ charge on Al. SiO_2 is a giant covalent network (like diamond) where every Si is tetrahedrally bonded to four O

atoms — all bonds must be broken to melt it. P_4O_{10}, SO_3, and Cl_2O_7 are simple molecular structures held by weak London forces (and dipole-dipole forces), so they have low melting points.

Exam Tip: When asked to explain melting point trends across Period 3 oxides, you must explain the change in structure type (giant ionic to giant covalent to simple molecular), not just say "bonding changes." The key is the type of forces that must be overcome: strong ionic/covalent bonds in giant structures vs weak intermolecular forces in simple molecular structures.

1.2 Reactions of Period 3 Oxides with Water

There is a clear trend in the acid-base behaviour of Period 3 oxides: metal oxides are basic (or amphoteric), while non-metal oxides are acidic.

Oxide	Reaction with water	Solution pH	Acid/base character
Na_2O	$Na_2O + H_2O \rightarrow 2NaOH$	~14 (strongly alkaline)	Basic
MgO	$MgO + H_2O \rightarrow Mg(OH)_2$ (sparingly soluble)	~9 (weakly alkaline)	Basic
Al_2O_3	Insoluble in water	—	Amphoteric
SiO_2	Insoluble in water (but reacts with hot conc. NaOH)	—	Weakly acidic
P_4O_{10}	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	~1 (strongly acidic)	Acidic
SO_3	$SO_3 + H_2O \rightarrow H_2SO_4$	~0 (strongly acidic)	Acidic
SO_2	$SO_2 + H_2O \rightarrow H_2SO_3$	~2 (acidic)	Acidic

Amphoteric oxide: An oxide that can react with both acids and bases. Al_2O_3 reacts with acids to form aluminium salts and with bases to form aluminate ions.

With acid: $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
With base: $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$

Why MgO gives a weaker alkaline solution than Na_2O: MgO is only sparingly soluble in water. The Mg^{2+} ion has a higher charge density than Na^+, creating stronger ionic bonds in the lattice that water cannot easily break. Therefore, fewer OH^- ions are produced.

Why SiO_2 does not react with water: SiO_2 has a giant covalent structure with very strong Si-O bonds throughout the lattice. Water cannot break these bonds. However, SiO_2 does react with hot concentrated NaOH or with molten NaOH:

$$SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$$

1.3 Reactions of Period 3 Oxides with Acids and Bases

Basic oxides react with acids:

$$Na_2O + 2HCl \rightarrow 2NaCl + H_2O$$

$$MgO + H_2SO_4 \rightarrow MgSO_4 + H_2O$$

Acidic oxides react with bases:

$$SO_3 + 2NaOH \rightarrow Na_2SO_4 + H_2O$$

$$P_4O_{10} + 12NaOH \rightarrow 4Na_3PO_4 + 6H_2O$$

1.4 Period 3 Chlorides

Chloride	NaCl	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_5
Bonding	Ionic	Ionic	Ionic solid; covalent vapour (Al_2Cl_6 dimers)	Simple covalent	Simple covalent (ionic solid: $PCl_4^+PCl_6^-$)
Structure	Giant ionic lattice	Giant ionic lattice	Giant ionic / molecular dimer	Simple molecular	Simple molecular / ionic solid
Reaction with water	Dissolves (neutral, pH 7)	Dissolves (slightly acidic, pH ~6.5)	Vigorous hydrolysis, acidic fumes of HCl	Vigorous hydrolysis, white fumes of HCl	Vigorous hydrolysis, white fumes of HCl

Ionic chlorides (NaCl, $MgCl_2$) dissolve in water because the ions are pulled apart by polar water molecules (ion-dipole interactions overcome lattice energy). NaCl gives a neutral solution (pH 7). $MgCl_2$ gives a slightly acidic

solution because the high charge density of Mg^{2+} attracts water molecules strongly, polarising the O-H bond and making it easier to release H^+:

$$[Mg(H_2O)_6]^{2+}(aq) \rightleftharpoons [Mg(H_2O)_5(OH)]^+(aq) + H^+(aq)$$

Covalent chlorides ($AlCl_3$, $SiCl_4$, PCl_5) undergo hydrolysis with water. The water molecule acts as a nucleophile, donating a lone pair to the δ+ atom. This breaks the M-Cl bond and produces HCl gas, which appears as white misty fumes in moist air:

$$AlCl_3(s) + 3H_2O(l) \rightarrow Al(OH)_3(s) + 3HCl(g)$$

$$SiCl_4(l) + 2H_2O(l) \rightarrow SiO_2(s) + 4HCl(g)$$

$$PCl_5(s) + 4H_2O(l) \rightarrow H_3PO_4(aq) + 5HCl(g)$$

Note: $SiCl_4$ produces SiO_2 rather than $Si(OH)_4$ because $Si(OH)_4$ is unstable and immediately loses water to form SiO_2.

Exam Tip: $AlCl_3$ is ionic in the solid state but behaves as a covalent compound when vaporised (forming Al_2Cl_6 dimers with coordinate bonds). Its reaction with water is hydrolysis, producing an acidic solution and fumes of HCl. Know the distinction between dissolving (ionic chlorides — ions separated) and hydrolysis (covalent chlorides — bonds broken, new substances formed). The white misty fumes of HCl are a key observation to state in exam answers.

2. Transition Metals

2.1 Definition and Electron Configuration

Transition metal: A d-block element that forms at least one stable ion with a partially filled d sub-shell. This definition excludes scandium (Sc^{3+} = [Ar] $3d^0$) and zinc (Zn^{2+} = [Ar] $3d^{10}$) because their ions have either empty or full d orbitals.

Key rule for electron configurations: When atoms form ions, the 4s electrons are removed first (despite being filled first in atoms) because once the 3d orbitals are occupied, the 4s orbital is at a higher energy. Then 3d electrons are removed if necessary.

Species	Electron configuration	Notes
Fe	[Ar] $3d^6$ $4s^2$	Standard filling
Fe^{2+}	[Ar] $3d^6$	Both 4s electrons removed first
Fe^{3+}	[Ar] $3d^5$	Half-filled d sub-shell (stable)
Cu	[Ar] $3d^{10}$ $4s^1$	Anomalous: one 4s electron promoted to complete 3d
Cu^+	[Ar] $3d^{10}$	Full d sub-shell (not a transition metal ion)
Cu^{2+}	[Ar] $3d^9$	Partially filled d sub-shell (transition metal ion)
Cr	[Ar] $3d^5$ $4s^1$	Anomalous: half-filled 3d is more stable
Cr^{3+}	[Ar] $3d^3$	Most common Cr ion
V	[Ar] $3d^3$ $4s^2$	Standard filling
Mn^{2+}	[Ar] $3d^5$	Half-filled, very stable

Exam Tip: Cr and Cu have anomalous electron configurations. Cr is [Ar] $3d^5$ $4s^1$ (not $3d^4$ $4s^2$) because a half-filled 3d sub-shell provides extra stability through exchange energy. Cu is [Ar] $3d^{10}$ $4s^1$ (not $3d^9$ $4s^2$) because a fully filled 3d sub-shell is more stable. These are very commonly examined.

2.2 Properties of Transition Metals

Transition metals have several characteristic properties, all arising from the partially filled d orbitals:

Property 1: Variable Oxidation States

Transition metals can form ions with different charges because the 3d and 4s electrons have similar energies, so different numbers of electrons can be removed without a dramatic increase in ionisation energy. This contrasts with s-block metals that typically only have one common oxidation state.

Metal	Common oxidation states	Key species
Ti	+2, +3, **+4**	TiO_2 (white pigment)

V	+2, +3, +4, **+5**	VO_2^+ (yellow), VO^{2+} (blue), V^{3+} (green), V^{2+} (violet)
Cr	+2, **+3**, +6	$Cr_2O_7^{2-}$ (orange), CrO_4^{2-} (yellow), Cr^{3+} (green)
Mn	**+2**, +4, +7	MnO_4^- (purple), MnO_2 (black solid)
Fe	**+2**, **+3**	Fe^{2+} (pale green), Fe^{3+} (yellow/brown)
Co	**+2**, +3	Co^{2+}(aq) (pink)
Ni	**+2**	Ni^{2+}(aq) (green)
Cu	+1, **+2**	Cu^{2+}(aq) (blue), Cu_2O (red)

The highest oxidation state for each first-row transition metal increases from Ti (+4) to Mn (+7) because more 3d electrons become available for bonding. After Mn, the maximum oxidation state decreases because the increasing nuclear charge holds the 3d electrons more tightly.

Property 2: Coloured Ions

Transition metal ions in solution are coloured because of the partially filled d orbitals. In an isolated (gaseous) ion, all five d orbitals have the same energy (they are degenerate). However, when ligands bond to the metal ion, the d orbitals split into two energy levels. This is explained by crystal field theory.

Electrons in the lower energy d orbitals can absorb photons of visible light with energy equal to the energy gap (Δ) and be promoted to the higher energy orbitals. The colour we observe is the complementary colour to the one absorbed.

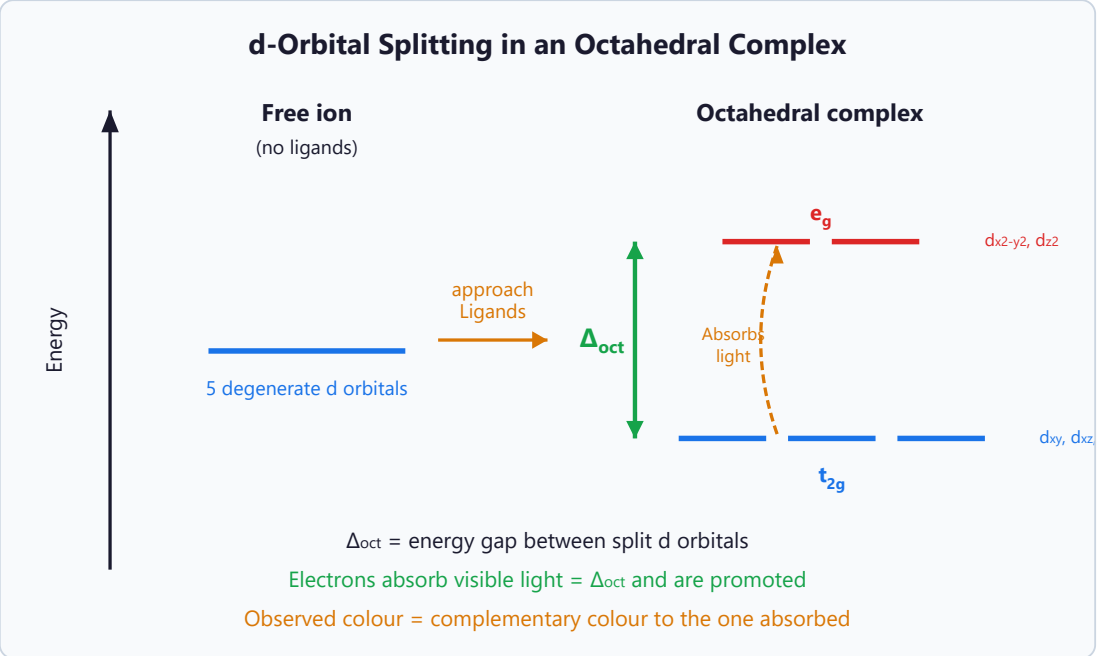
Ion	Oxidation state	Colour in aqueous solution	Colour absorbed
Cu^{2+}(aq)	+2	Blue	Red/orange
Fe^{2+}(aq)	+2	Pale green	Red
Fe^{3+}(aq)	+3	Yellow/brown	Violet/blue
Cr^{3+}(aq)	+3	Green (ruby-red in some complexes)	Red (or green)
Mn^{2+}(aq)	+2	Very pale pink (almost colourless)	Green

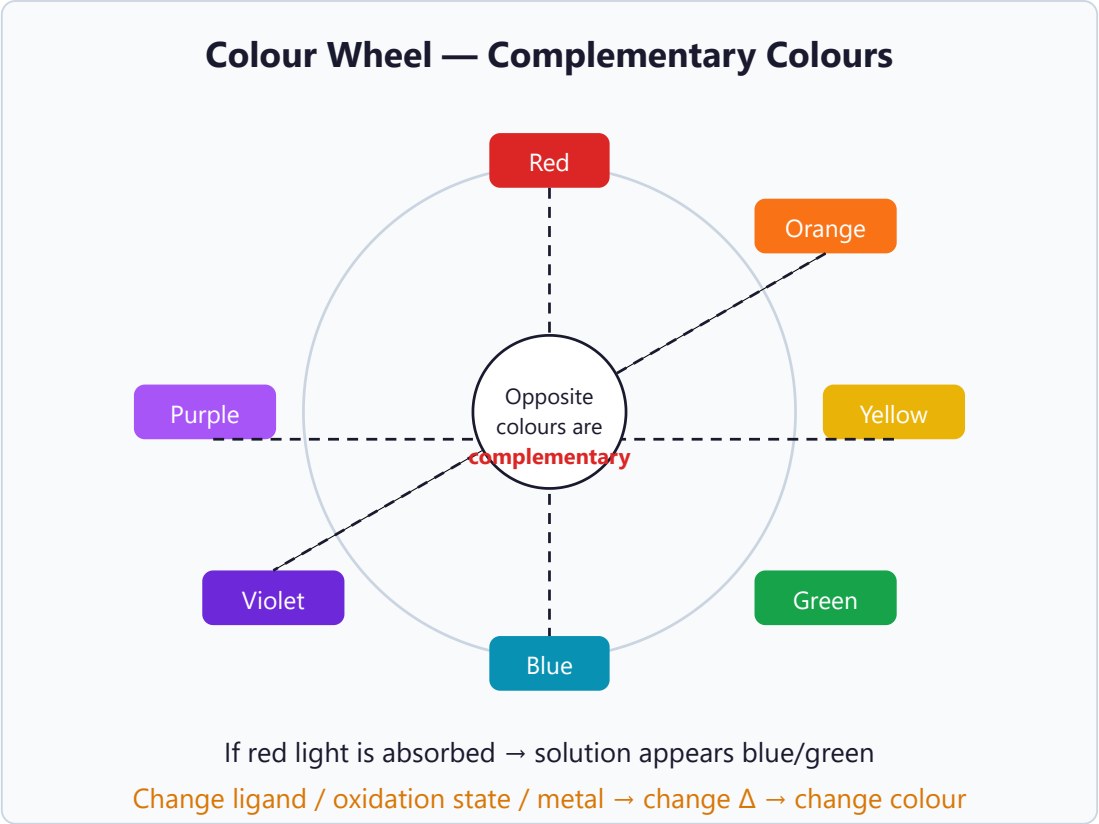
$Co^{2+}(aq)$	+2	Pink	Green
$Ni^{2+}(aq)$	+2	Green	Red
$MnO_4^-(aq)$	+7	Purple	Green/yellow
$CrO_4^{2-}(aq)$	+6	Yellow	Violet
$Cr_2O_7^{2-}(aq)$	+6	Orange	Blue

Factors that affect colour:

- **The identity of the metal ion** (different metals have different d orbital energies)
- **The oxidation state** (higher charge = stronger interaction with ligands = larger Δ)
- **The ligands present** (different ligands cause different degrees of splitting)
- **The coordination number** (affects the geometry and hence the splitting pattern)

Why Sc^{3+} and Zn^{2+} are colourless: Sc^{3+} has no d electrons ($3d^0$) so there are no electrons to be promoted. Zn^{2+} has a full d sub-shell ($3d^{10}$) so there are no empty d orbitals available for promotion.





Property 3: Catalytic Behaviour

Transition metals and their compounds are effective catalysts because:

- **Variable oxidation states** allow them to transfer electrons to/from reactants, providing an alternative reaction pathway with lower activation energy (homogeneous catalysis).
- **Surface adsorption:** Reactant molecules can bind to the surface of the metal (heterogeneous catalysis), weakening bonds in the reactants and holding them in the correct orientation to react.
- The availability of empty or partially filled d orbitals allows bonding with a wide range of species.

Catalyst	Process	Type	Industrial significance
Fe	Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$	Heterogeneous	Ammonia for fertilisers
V_2O_5	Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$	Heterogeneous	Sulfuric acid production
MnO_2	Decomposition of H_2O_2	Heterogeneous	Laboratory preparation of O_2
Ni	Hydrogenation of alkenes (C=C + H_2)	Heterogeneous	Margarine production
Fe^{3+}/Fe^{2+}	$S_2O_8^{2-} + 2I^-$ reaction	Homogeneous	Classic A-level example

Mn^{2+} (autocatalysis)	$MnO_4^- + C_2O_4^{2-}$	Homogeneous	Titration speeds up as it proceeds

Heterogeneous catalyst: A catalyst in a different phase from the reactants (usually a solid catalyst with gaseous or liquid reactants). Reactants adsorb onto the surface, react, and products desorb.

Homogeneous catalyst: A catalyst in the same phase as the reactants (usually all in aqueous solution). It works by changing oxidation state during the reaction, being regenerated at the end.

Example of homogeneous catalysis — Fe^{3+} catalysing the reaction between $S_2O_8^{2-}$ and I^-:

The overall reaction is: $S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$

This is slow because both reactants are negatively charged and repel each other. Fe^{3+} provides an alternative pathway:

$$\text{Step 1: } 2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$$

$$\text{Step 2: } 2Fe^{2+}(aq) + S_2O_8^{2-}(aq) \rightarrow 2Fe^{3+}(aq) + 2SO_4^{2-}(aq)$$

The Fe^{3+} is regenerated — it is not consumed overall. Each step involves a positive ion reacting with a negative ion, reducing electrostatic repulsion and lowering activation energy.

Property 4: Formation of Complex Ions

See section 2.3 below.

2.3 Complex Ions

Complex ion: A central metal ion surrounded by ligands. A **ligand** is a molecule or ion that donates a lone pair of electrons to the central metal ion to form a dative (coordinate) covalent bond. The **coordination number** is the total number of dative bonds from all ligands to the central metal ion.

Types of ligands:

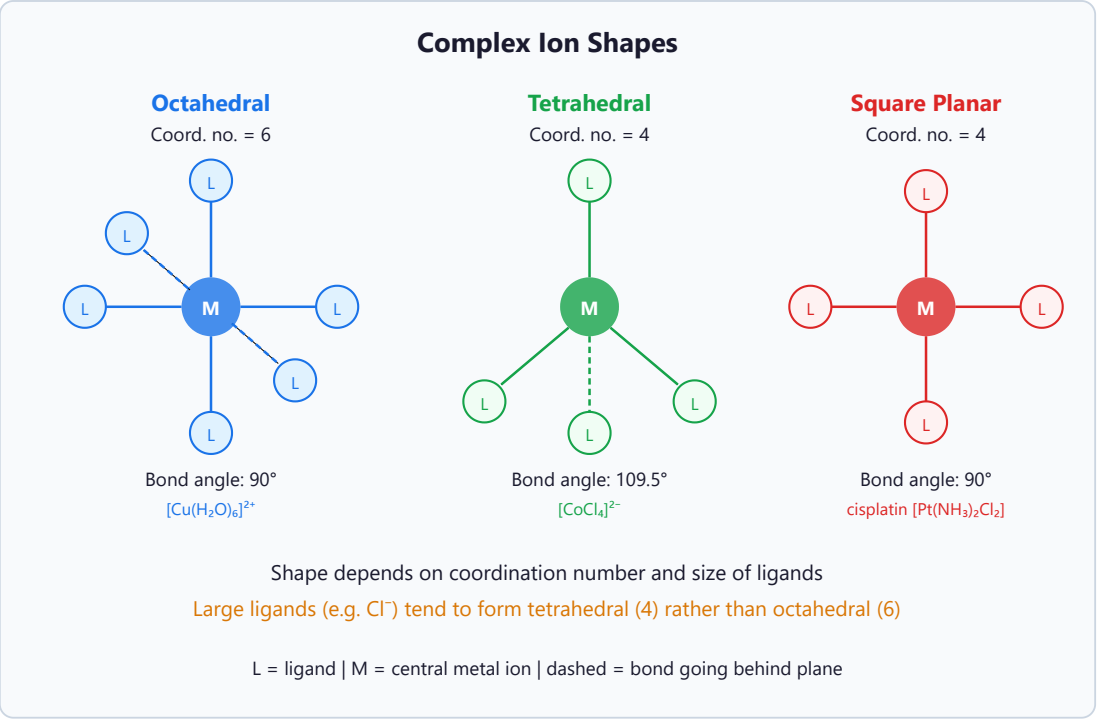
Type	Description	Examples	Coordination number contribution per ligand

Monodentate	Forms one dative bond (has one lone pair that bonds to the metal)	H_2O, NH_3, Cl^-, CN^-, OH^-, CO	1
Bidentate	Forms two dative bonds per ligand (has two lone pairs that can bond)	Ethane-1,2-diamine (en, $H_2NCH_2CH_2NH_2$), ethanedioate ($C_2O_4^{2-}$)	2
Multidentate (polydentate)	Forms multiple dative bonds per ligand	$EDTA^{4-}$ (hexadentate, forms 6 bonds), haem (tetradentate)	4 or 6

Note on H_2O vs NH_3: Both have lone pairs and are similar in size, so they are interchangeable in many complexes. NH_3 is a slightly stronger ligand than H_2O (it causes a larger d-orbital splitting), which is why ammonia complexes are often a deeper colour.

Common shapes of complexes:

- **Coordination number 6 → Octahedral** (bond angle 90°): e.g. $[Cu(H_2O)_6]^{2+}$, $[Fe(CN)_6]^{3-}$, $[Cr(NH_3)_6]^{3+}$, $[Co(NH_3)_6]^{3+}$
- **Coordination number 4 → Tetrahedral** (bond angle 109.5°): e.g. $[CoCl_4]^{2-}$, $[CuCl_4]^{2-}$. Usually formed with large ligands like Cl^- (too bulky for 6 to fit).
- **Coordination number 4 → Square planar** (bond angle 90°): e.g. cisplatin $[PtCl_2(NH_3)_2]$, $[NiCl_2(NH_3)_2]$. Less common; usually with Pt^{2+}, Pd^{2+}, Ni^{2+}.
- **Coordination number 2 → Linear** (bond angle 180°): e.g. $[Ag(NH_3)_2]^+$ (Tollens' reagent), $[CuCl_2]^-$



2.4 Isomerism in Transition Metal Complexes

Complexes can exhibit two types of stereoisomerism:

Cis-trans isomerism occurs in square planar and octahedral complexes with two types of ligand:

- **Cis:** identical ligands are adjacent (next to each other)
- **Trans:** identical ligands are opposite each other

Example: $[Pt(NH_3)_2Cl_2]$ (square planar) has cis and trans forms. Only the cis form (cisplatin) is an effective anticancer drug.

Optical isomerism occurs in octahedral complexes with bidentate or multidentate ligands:

- $[Cr(en)_3]^{3+}$ has two non-superimposable mirror images (enantiomers)
- $[Cr(C_2O_4)_3]^{3-}$ also exhibits optical isomerism
- The two enantiomers rotate plane-polarised light in opposite directions

Exam Tip: Optical isomerism in complexes only occurs with bidentate/multidentate ligands in octahedral complexes, not with monodentate ligands. You need to be able to draw the two mirror-image forms. The complex must have no plane of symmetry.

2.5 Ligand Substitution Reactions

Ligands can be exchanged (substituted) in complex ions. The position of equilibrium depends on the relative strength of the ligand-metal bonds and entropy changes.

Copper(II) complexes

$$[Cu(H_2O)_6]^{2+}(aq) + 4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4(H_2O)_2]^{2+}(aq) + 4H_2O(l)$$

Pale blue solution → deep blue solution. Only 4 of the 6 water ligands are replaced by ammonia (partial substitution). The coordination number remains 6 and the shape remains octahedral. This is because H_2O and NH_3 are similar in size.

$$[Cu(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 6H_2O(l)$$

Pale blue solution → yellow/green solution. The coordination number changes from 6 to 4 and the shape changes from octahedral to tetrahedral because Cl^- is a much larger ligand than H_2O — only 4 Cl^- ions can fit around the Cu^{2+} ion.

Cobalt(II) complexes

$$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq) + 6H_2O(l)$$

Pink solution → blue solution. Octahedral (6) → tetrahedral (4). This is the basis of cobalt chloride paper: the paper is blue when dry ($[CoCl_4]^{2-}$) and turns pink when wet ($[Co(H_2O)_6]^{2+}$).

Chromium(III) complexes

$$[Cr(H_2O)_6]^{3+}(aq) + 3NH_3(aq) \rightarrow Cr(OH)_3(s) + 3NH_4^+(aq) \text{ (initially)}$$

$$Cr(OH)_3(s) + 6NH_3(aq) \rightarrow [Cr(NH_3)_6]^{3+}(aq) + 3OH^-(aq) \text{ (in excess)}$$

Green/violet solution → green precipitate → purple solution in excess ammonia.

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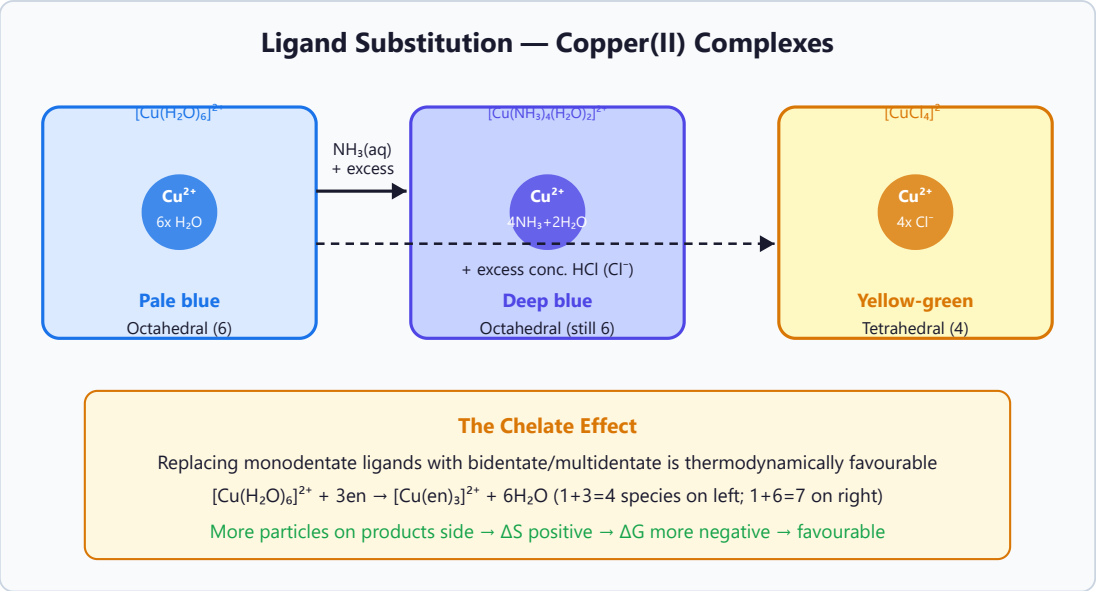


The Chelate Effect Explained

When monodentate ligands are replaced by bidentate or multidentate ligands, the reaction is thermodynamically favourable because of an increase in entropy. For example:

$$[Cu(H_2O)_6]^{2+}(aq) + 3en(aq) \rightarrow [Cu(en)_3]^{2+}(aq) + 6H_2O(l)$$

On the left, there are 4 species in solution (1 complex + 3 en). On the right, there are 7 species (1 complex + 6 H_2O). The increase in the number of particles leads to a positive entropy change ($\Delta S > 0$). Since $\Delta G = \Delta H - T\Delta S$, a positive ΔS makes ΔG more negative, driving the reaction forward. The enthalpy change is approximately zero (because the M-N bond strengths are similar whether from NH_3 or en), so the entropy term dominates.

Exam Tip: When describing ligand substitution, always state: (1) the change in colour, (2) any change in coordination number, (3) any change in shape, (4) whether the substitution is partial or complete. Write the full balanced equation with square brackets and charges. For chelate effect questions, explain the entropy increase in terms of the number of particles in solution.

3. Reactions of Ions in Aqueous Solution

3.1 Tests for Metal Cations Using NaOH(aq)

Adding dilute sodium hydroxide solution to aqueous metal ions produces coloured precipitates of metal hydroxides. These form because the OH^- acts as a ligand, replacing water ligands:

$$[M(H_2O)_6]^{n+}(aq) + nOH^-(aq) \rightarrow M(OH)_n(s) + 6H_2O(l)$$

Ion	Colour of precipitate with NaOH(aq)	Effect of excess NaOH(aq)
Cu^{2+}	Blue precipitate $Cu(OH)_2$	Insoluble (no change)
Fe^{2+}	Green precipitate $Fe(OH)_2$	Insoluble; slowly turns brown in air (oxidised to $Fe(OH)_3$ by O_2)
Fe^{3+}	Brown precipitate $Fe(OH)_3$	Insoluble (no change)
Al^{3+}	White precipitate $Al(OH)_3$	Dissolves in excess (amphoteric): $Al(OH)_3 + OH^- \rightarrow [Al(OH)_4]^-$
Cr^{3+}	Green precipitate $Cr(OH)_3$	Dissolves in excess (amphoteric): $Cr(OH)_3 + 3OH^- \rightarrow [Cr(OH)_6]^{3-}$
Mn^{2+}	Pale brown/buff precipitate $Mn(OH)_2$	Insoluble; darkens in air (oxidised)

3.2 Tests for Metal Cations Using NH_3(aq)

Adding dilute ammonia solution produces similar precipitates initially (NH_3 acts as a weak base, releasing OH^-). However, with excess NH_3, some precipitates dissolve because NH_3 can act as a ligand:

Ion	With a few drops of NH_3(aq)	With excess NH_3(aq)
Cu^{2+}	Blue precipitate	Dissolves to deep blue solution: $[Cu(NH_3)_4(H_2O)_2]^{2+}$
Fe^{2+}	Green precipitate	Insoluble (no change)
Fe^{3+}	Brown precipitate	Insoluble (no change)
Al^{3+}	White precipitate	Insoluble (no change) — distinguishes from Cr^{3+}
Cr^{3+}	Green precipitate	Dissolves to purple solution: $[Cr(NH_3)_6]^{3+}$
Co^{2+}	Blue-green precipitate	Dissolves to straw/brown solution: $[Co(NH_3)_6]^{2+}$

Exam Tip: Use the difference between NaOH and NH_3 tests to distinguish between Al^{3+} and Cr^{3+}. Both give precipitates that dissolve in excess NaOH (amphoteric). But with excess NH_3: $Cr(OH)_3$ dissolves (NH_3 is

a strong enough ligand) while $Al(OH)_3$ does not dissolve (NH_3 cannot displace OH^- from Al^{3+} because Al^{3+} is not a transition metal and its small size prevents 6 NH_3 molecules from fitting around it).

3.3 Tests for Anions

Anion	Test	Positive result	Equation
CO_3^{2-}	Add dilute acid	Effervescence; gas turns limewater milky	$CO_3^{2-} + 2H^+ \rightarrow H_2O + CO_2$
SO_4^{2-}	Add dilute HCl, then $BaCl_2(aq)$	White precipitate of $BaSO_4$ (insoluble in excess HCl)	$Ba^{2+} + SO_4^{2-} \rightarrow BaSO_4$
Cl^-	Add dilute HNO_3, then $AgNO_3(aq)$	White precipitate (AgCl); soluble in dilute NH_3	$Ag^+ + Cl^- \rightarrow AgCl$
Br^-	Add dilute HNO_3, then $AgNO_3(aq)$	Cream precipitate (AgBr); soluble in conc. NH_3	$Ag^+ + Br^- \rightarrow AgBr$
I^-	Add dilute HNO_3, then $AgNO_3(aq)$	Yellow precipitate (AgI); insoluble in NH_3	$Ag^+ + I^- \rightarrow AgI$
NH_4^+	Warm with NaOH(aq)	Pungent gas turns damp red litmus blue (NH_3)	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$

Exam Tip: For the sulfate test, add HCl first to remove carbonates (CO_3^{2-}) and sulfites (SO_3^{2-}) that would also give white precipitates with barium ions. Do NOT use sulfuric acid (which adds SO_4^{2-}). For the halide test, use nitric acid (not HCl, which adds Cl^-). The solubility of silver halides in ammonia helps distinguish them: AgCl dissolves in dilute NH_3, AgBr needs concentrated NH_3, AgI does not dissolve at all.

3.4 Acidity of Transition Metal Aqua Ions

Transition metal aqua ions such as $[Fe(H_2O)_6]^{3+}$ are acidic in solution. The highly charged, small metal ion has a high charge density, which strongly attracts the electrons in the O-H bonds of the coordinated water molecules. This weakens the O-H bond, making it easier for the water to lose a proton (H^+) to a surrounding free water molecule (hydrolysis/deprotonation):

$$[Fe(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Fe(H_2O)_5(OH)]^{2+}(aq) + H_3O^+(aq)$$

Key principle: M^{3+} ions are more acidic than M^{2+} ions because the higher charge and smaller radius give a greater charge density, which polarises the O-H bonds in coordinated water more strongly. For example, $[Fe(H_2O)_6]^{3+}$ solutions have pH ~3, while $[Fe(H_2O)_6]^{2+}$ solutions have pH ~5.

This acidity explains why adding Na_2CO_3 to $Fe^{3+}(aq)$ does not produce $FeCO_3$ but instead produces $Fe(OH)_3$ precipitate with CO_2 effervescence — the Fe^{3+} is too acidic to coexist with carbonate ions.

3.5 Redox Titrations Using Potassium Manganate(VII)

$KMnO_4$ is widely used as an oxidising agent in redox titrations because it acts as its own indicator — the purple MnO_4^- is decolourised as it is reduced to the almost colourless Mn^{2+}:

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(l) \quad (E° = +1.52\ V)$$

Common titration with iron(II):

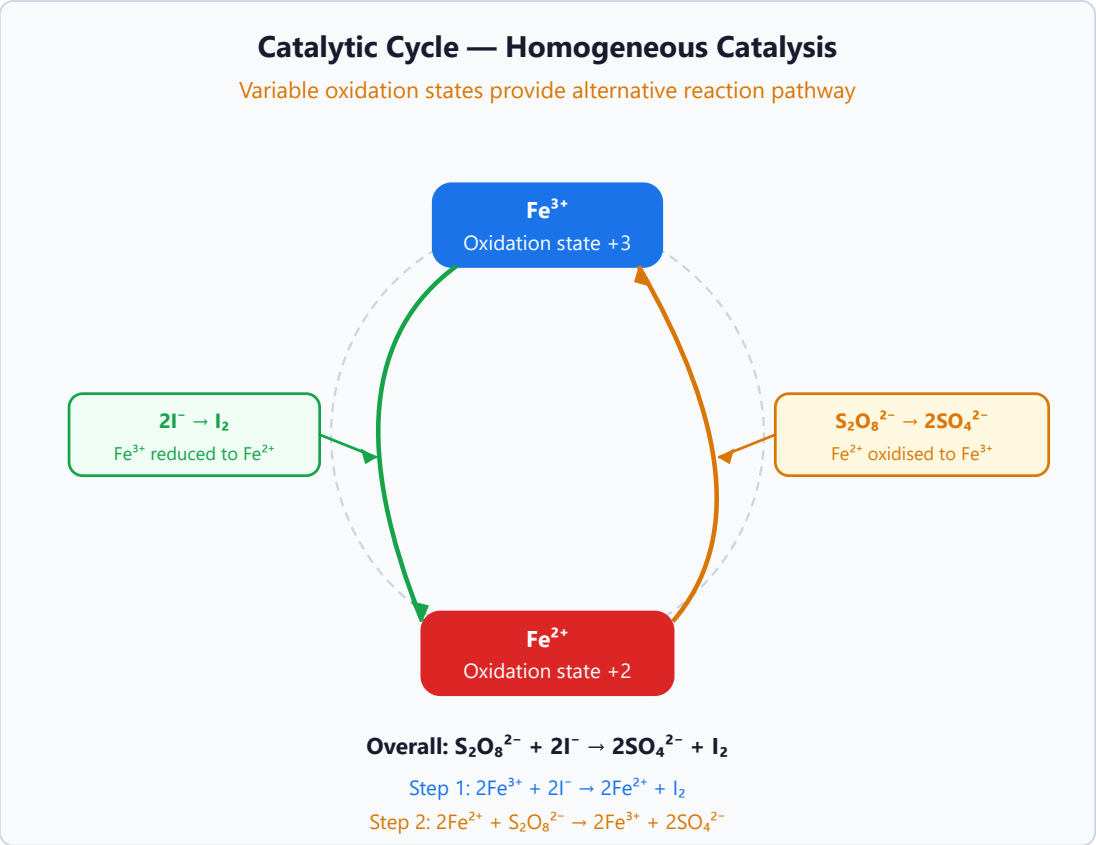
$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

The solution is acidified with dilute H_2SO_4 (not HCl, which would be oxidised by MnO_4^-). The end point is when the first drop of $KMnO_4$ gives a persistent pink/purple colour that does not fade after 30 seconds.

Common titration with ethanedioate (oxalate):

$$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$$

This must be heated to ~60°C. It demonstrates autocatalysis: the reaction starts slowly because there is no catalyst, but as Mn^{2+} forms, it catalyses the reaction, so it speeds up (the purple colour fades more quickly as the titration proceeds).



3.6 Summary of Key Colour Changes

Reaction / Species	Colour change
$[Cu(H_2O)_6]^{2+}$ + excess NH_3	Pale blue → deep blue
$[Cu(H_2O)_6]^{2+}$ + excess Cl^-	Blue → yellow/green
$[Co(H_2O)_6]^{2+}$ + excess Cl^-	Pink → blue
$[Fe(H_2O)_6]^{2+}$ in air	Pale green → yellow/brown (oxidation)
MnO_4^- + Fe^{2+}	Purple → colourless (Mn^{2+}) + yellow (Fe^{3+})
$Cr_2O_7^{2-}$ + reducing agent	Orange → green (Cr^{3+})
$V^{5+} \rightarrow V^{4+} \rightarrow V^{3+} \rightarrow V^{2+}$	Yellow → blue → green → violet

Exam Tip: Colour changes are among the most frequently examined topics. You must be able to state both the initial and final colours, write the full equation, and explain the colour change in terms of d-orbital

splitting. Questions often ask you to use electrode potentials to predict whether a reaction will occur, then describe the expected colour change.